

SODS Final Project - Group 12

Lorenzo Pavone s362884 Samuele Martello s360819 Matteo Scardovi s352118

1 Baseline Synthesis

Table 1: Baseline synthesis results (clock period = 3.0 ns, STcmos65 HVT, nominal 1.20 V, 25°C)

Metric	Value
Clock period (ns)	3.0
Area (μm^2)	39130.95
Slack (ns)	0.000013
Leakage power (mW)	8.04×10^{-8}
Dynamic power (mW)	6.04×10^{-3}

The minimum clock period, among multiples of 0.5 ns, satisfying both the timing (slack ≥ 0) and the area (area $< 39200 \mu m^2$) constraints is **3.0 ns** (333 MHz). The constraints were verified in PrimeTime by sweeping the clock period downward: at 4.5, 4.0 and 3.5 ns both constraints are met with increasing margin reduction, at 3.0 ns the design still closes with a positive slack and an area of $39130.95 \mu m^2$ (just below the $39200 \mu m^2$ limit), whereas at 2.5 ns the area grows beyond $42000 \mu m^2$, violating the area constraint. The design was synthesised using only HVT cells under nominal operating conditions. Power values were obtained with VCD-based switching-activity back-annotation in PrimeTime from a post-synthesis functional simulation.

2 Power Optimization

2.1 Clock-Gating

Table 2: Clock-gating optimization results (minimum bitwidth = 8, maximum fanout = 32)

Metric	Value
Area (μm^2)	37602.04
Slack (ns)	0.000022
Leakage power (mW)	8.03×10^{-8}
Dynamic power (mW)	6.20×10^{-3}

Discussion. Clock gating inserts Integrated Clock-Gating (ICG) cells that block the clock signal to groups of registers whenever their value does not need to change, reducing the number of clock transitions and therefore the dynamic power associated with the clock network.

In this experiment the area **decreases** by approximately 4 % compared to the baseline (37602 vs. $39131 \mu m^2$). Although each ICG cell adds some area, this is more than compensated by the removal of the feedback multiplexers that, in the baseline, were used to recirculate the value of each enabled register: by gating the clock instead, those per-bit multiplexers are no longer required, and the overall cell count decreases. The leakage power is essentially unchanged with respect to the baseline (8.03×10^{-8} vs. 8.04×10^{-8} mW), since the design still uses HVT cells only and the total transistor count is comparable.

The dynamic power measured here is slightly **higher** than in the baseline (6.20 vs. 6.04 mW). This counter-intuitive result is a consequence of the short simulation used to generate the VCD: the testbench covers only a few SHA-256 hash computations, so the clock-gating cells are toggled frequently (they switch on and off at every new data block) rather than holding the clock low for long idle intervals. In a real workload where the SHA-256 core spends most of its time idle, clock gating would significantly reduce dynamic power. The slack remains positive and very close to zero, confirming that the clock-gated design still meets the 3.0 ns timing constraint.

2.2 Clock-Gating + Multi-VT

Table 3: Clock-gating and Multi-VT optimization results

Metric	Value
Percentage of LVT cells (%)	31.60
Percentage of SVT cells (%)	23.64
Percentage of HVT cells (%)	44.76
Area (μm^2)	31828.96
Slack (ns)	0.000147
Leakage power (mW)	4.57×10^{-6}
Dynamic power (mW)	4.81×10^{-3}

Discussion. Multi-threshold voltage (Multi-VT) synthesis allows Design Compiler to choose from three cell families — LVT (low threshold voltage), SVT (standard), and HVT (high) — for each gate individually. LVT cells switch faster but leak more; HVT cells are slower but have minimal leakage. DC exploits this freedom to meet the tight 3.0 ns timing constraint at minimum cost.

The resulting cell mix reflects the timing pressure: about 45 % HVT cells are kept on non-critical paths where leakage is the priority, while 32 % LVT cells are placed on the critical paths to provide the required speed (slack ≈ 0), and the remaining 24 % SVT cells fill the intermediate paths where a balanced trade-off between speed and leakage is preferable.

Compared to clock-gating alone, the area **decreases** by about 15 % (31829 vs. 37602 μm^2) because, at the same drive strength, the faster LVT and SVT cells let DC meet timing with smaller gates on the critical paths. The leakage power **increases** by almost two orders of magnitude relative to the baseline (4.57×10^{-6} vs. 8.04×10^{-8} mW) due to the significant fraction of LVT and SVT cells, which have higher sub-threshold leakage current than HVT cells. The dynamic power **decreases** with respect to clock-gating alone (4.81 vs. 6.20 mW) because the smaller cell sizes reduce node capacitances. The slack stays positive and close to zero (0.000147 ns), confirming that DC has pushed the design to its performance limit under the given constraints.